

CASE 29

INTERLEUKIN 1 RECEPTOR-ASSOCIATED KINASE 4 DEFICIENCY

An inability of the innate immune system to detect the presence of pathogens.

The presence of an invading microorganism is first recognized by tissue macrophages at the site of infection. Pathogens carry characteristic chemical structures on their surface and in their nucleic acids that are not present in human cells. These pathogen-associated molecular patterns (PAMPs) are recognized by invariant pattern-recognition receptors (PRRs) on macrophages, and also on neutrophils, which take up the pathogen and destroy it. PRRs are also present on dendritic cells, whose function is to present antigen to and activate naive T cells. Dendritic cells form the link between the innate and the adaptive immune response. Among the PRRs are the Toll-like receptors (TLRs), which are present on the cell surface and in the membranes of endosomes, and which enable cells of the innate immune system to detect and respond to a wide variety of pathogens.

There are 10 different TLRs in humans, which between them recognize microbial nucleic acids, such as unmethylated bacterial DNA and long double-stranded RNA, as well as molecules specific to particular classes of microorganisms, such as the bacterial lipopolysaccharide (LPS or endotoxin) characteristic of Gram-negative bacteria and the protein flagellin of bacterial flagella (Fig. 29.1). All these molecules are essential for the microorganism's survival and are therefore relatively invariant. The limited number of TLRs can thus detect the presence of many different microorganisms by recognizing features that are typical of the different groups of pathogens—bacteria, viruses, fungi, and parasites.

The TLRs are signaling receptors, and their activation leads to enhancement of the antimicrobial activity of macrophages and neutrophils, and the secretion of cytokines by macrophages, which help attract more macrophages and neutrophils out of the blood and into the site of infection. Upon engagement of their TLRs, dendritic cells undergo a brief period of macropinocytosis that is important for taking up and processing microbial antigens for subsequent presentation to naive T cells. Signaling via TLRs also induces the maturation of tissue dendritic cells and their migration to peripheral lymphoid tissues such as lymph nodes, where they encounter and activate antigen-specific T cells. During maturation, conventional dendritic cells upregulate the production of co-stimulatory molecules such as CD40, B7.1 (CD80), and B7.2 (CD86), which are essential for T-cell activation.

This case was prepared by Raif Geha, MD, in collaboration with Douglas McDonald, MD, PhD.

TOPICS BEARING ON THIS CASE:

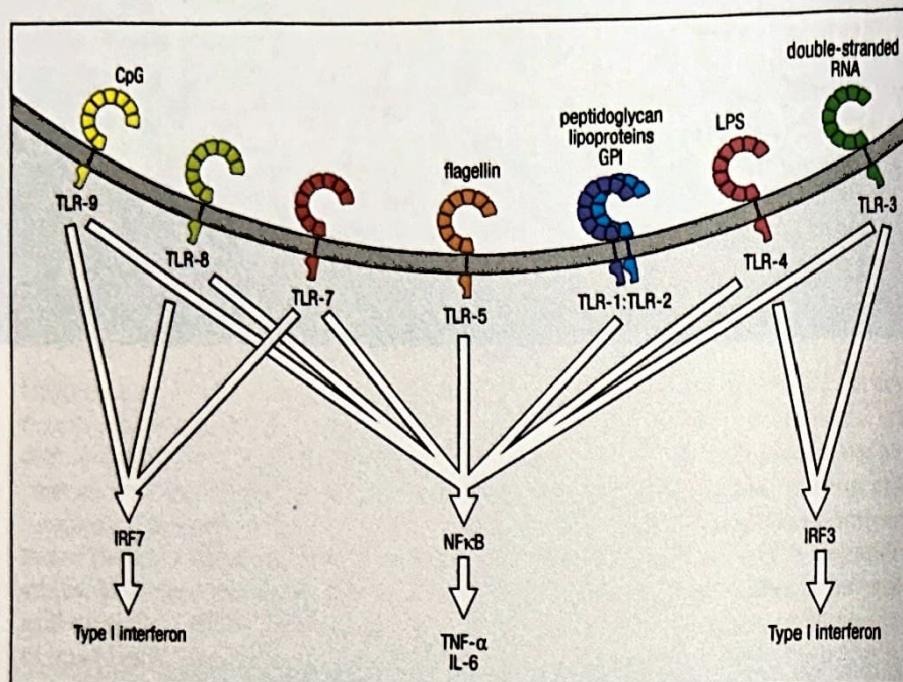
Innate immunity

Toll-like receptors

Toll-like receptor/
IL-1R signaling
pathways

Dendritic cells and
the initiation of
adaptive immune
responses

Fig 29.1 Toll-like receptors and cytokine production. Toll-like receptors (TLRs) and their representative ligands are shown. Simplified signaling pathways are shown below each receptor, leading to the production of cytokines characteristic of each TLR. TLR-7, TLR-8, and TLR-9 are present in the membranes of endosomes, where they recognize ligands generated in the endosomal pathway from microbes engulfed by phagocytosis or macropinocytosis. The other TLRs are expressed on the cell surface. GPI, glycosylphosphatidylinositol (*Trypanosoma cruzi*); CpG, unmethylated DNA; LPS, bacterial lipopolysaccharide.



They also produce chemokines and cytokines that help induce adaptive immune responses, such as the pro-inflammatory cytokines tumor necrosis factor- α (TNF- α), interleukin-1 (IL-1), IL-6, and IL-12 (see Case 23). Another type of dendritic cell, the plasmacytoid dendritic cell, produces large amounts of the antiviral type I interferons (IFN- α and IFN- β) in response to viral infections.

The stimulation of dendritic cells by PAMPs such as LPS and unmethylated CpG in bacterial DNA is particularly critical for their ability to induce the development of T_H1 effector cells. The exposure of dendritic cells to pro-inflammatory cytokines is sufficient to enable them to induce the proliferation (clonal expansion) of antigen-specific naive T cells, but the upregulation of co-stimulatory molecules and of IL-12 production as a result of TLR stimulation is essential for dendritic cells to be able to induce the differentiation of T_H1 effector functions, such as the secretion of IFN- γ , in CD4 T cells. Thus, TLRs have a vital role in the development of both innate and adaptive immune responses.

The TLRs and the IL-1 receptor signal through a common intracellular signaling pathway, and this case shows how a defect in this pathway has serious effects on a person's ability to respond to some classes of pathogens.

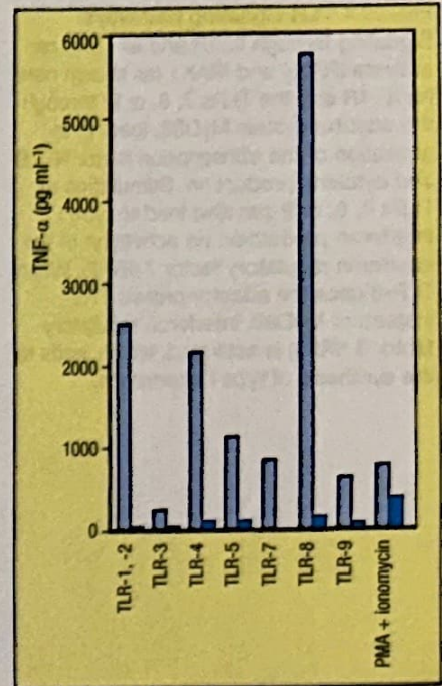
The case of Douglas Mooster: a 6-year-old boy with recurrent pneumococcal meningitis.

Douglas is an only child and was referred to the Immunology clinic at the Boston Children's Hospital at the age of 6 years after two episodes of pneumococcal meningitis (the first a year previously) caused by serotypes 3 and 14, respectively. As a result of the meningitis, Douglas had suffered a stroke and seizures, and had been left with mild hearing loss and mild learning impairment. When he was first seen at the immunology clinic, his overall neurological function was good.

Frequent middle-ear infections (otitis media) were a significant part of Douglas's medical history, and myringotomy tubes had been inserted in both ears when he was 2 years old to aerate the middle ear. At 11 months old he had developed intestinal intussusceptions (telescoping of a segment of the intestine into an adjacent segment), complicated by perforation of the intestine and formation of an abscess in

Recurrent pyogenic infections.
No fever.

Fig. 29.2 Impaired TLR-induced TNF- α production in IRAK4 deficiency. Peripheral blood mononuclear cells from Douglas (dark blue) and a normal control (light blue) were stimulated by ligands for the TLR receptors indicated or by phorbol myristate acetate (PMA) plus ionomycin as a positive control. The latter compounds bypass receptor signaling by activating protein kinase C and causing an influx of calcium, respectively. TNF- α production was measured from cell supernatants 12 hours after stimulation. The histogram shows that Douglas's cells produced no or very little TNF- α in comparison with the response of normal cells to TLR ligands.



the peritoneum. He had also suffered from boils (furuncles) of the scalp and several skin infections, all of which had responded well to treatment with antibiotics. His medical history also noted an unusually weak febrile response, as Douglas usually developed only low-grade fevers late in the course of these illnesses. He seemed to have no particular susceptibility to viral infections.

On physical examination, he appeared thin but otherwise well. His tonsils were slightly enlarged, as were the lymph nodes of the anterior cervical (neck) chains (lymphadenopathy). His abdomen was soft and not tender, although the liver could be felt about 3 cm below the lower rib on the right (the liver is normally non-palpable). The spleen was not enlarged and his skin showed no rash or other unusual features.

Laboratory tests showed a normal complete blood count, complement function, and immunoglobulin titer. Specific antibody titers showed that Douglas had made protective immune responses to protein antigens, such as tetanus toxoid, after his routine immunizations. Douglas had been given the vaccine Pneumovax as part of the evaluation at the immunology clinic; this vaccine contains polysaccharide antigens from 23 different serotypes of *Streptococcus pneumoniae* (pneumococcus), and he had not made responses to any of the 14 serotypes that were tested for at the clinic. He had also failed to mount a protective antibody titer after immunization with polysaccharide antigens from *Neisseria meningitidis* (meningococcus), a common cause of meningitis. He had, however, normal absolute numbers of B cells and T cells.

Given Douglas's susceptibility to pneumococcal infections, with no obvious underlying immunological cause, and his poor febrile response to infection, his Toll-like receptor (TLR) function was evaluated. A sample of peripheral blood cells (PBCs) produced little or no TNF- α in response to known TLR ligands when tested in the laboratory (Fig. 29.2). The integrity of the signal transduction pathways leading from the TLRs was evaluated *in vitro* by testing the cells' response to IL-1, as the IL-1 receptor (IL-1R) stimulates the same signaling pathway. Stimulation of the cells with IL-1 produced no activation of MAP kinases and no phosphorylation of I κ B α , important intermediate steps in these pathways (Fig. 29.3). These results were consistent with a defect very early in the pathway, and a defect in the IL-1R-associated kinase 4 (IRAK4) was suspected. Sequencing of Douglas's DNA for the *IRAK4* gene revealed a homozygous nonsense mutation within the kinase domain, resulting in the production of a truncated protein with no kinase activity.

Adaptive immunity mostly normal. Test for TLR function.

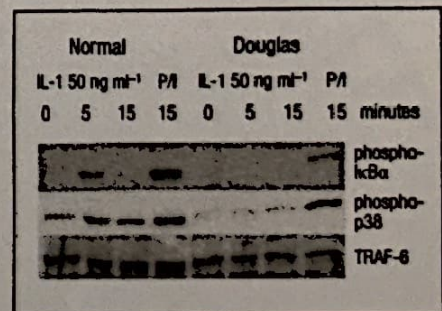


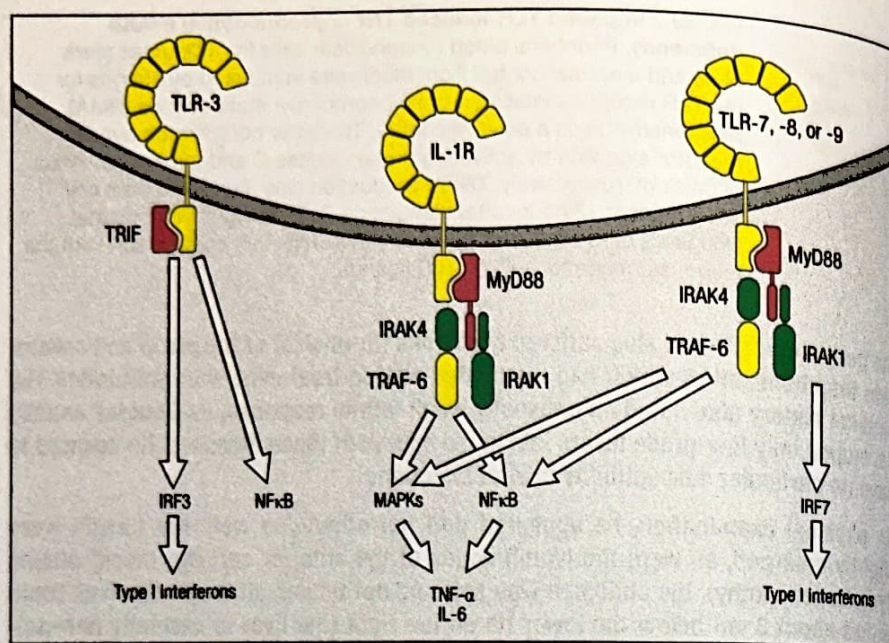
Fig. 29.3 Impaired IL-1-induced phosphorylation of p38 MAP kinase and I κ B α in IRAK4 deficiency. Peripheral blood mononuclear cells from Douglas and a normal control were stimulated with IL-1 or PMA plus ionomycin (P/I) as a positive control for the times indicated. Cell lysates were subjected to Western blotting with antibodies against phosphorylated p38 MAP kinase or phosphorylated I κ B α . No phospho-I κ B α or phospho-p38 were present in Douglas's blood cells. Antibody against TRAF-6 is shown as a protein-loading control. Courtesy of D. McDonald.

Interleukin 1 receptor-associated kinase 4 deficiency (IRAK4 deficiency).

The IL-1 receptor (IL-1R) and the TLRs share a common signal transduction pathway that involves the adaptor protein MyD88, the signaling intermediate TRAF-6, and the receptor-associated protein kinases IRAK1 and IRAK4. Activation of these receptors leads to recruitment of MyD88 to the receptor, followed by the recruitment

Fig. 29.4 TLR signaling pathways.

Signaling through IL-1R and all TLRs can activate IRAK4 and IRAK1 (as shown here for IL-1R and the TLRs 7, 8, or 9) through the adaptor protein MyD88, leading to activation of the transcription factor NF κ B and cytokine production. Stimulation of TLRs 7, 8, or 9 can also lead to type I interferon production via activation of the interferon regulatory factor 7 (IRF7). When TLR-3 uses the adaptor protein TRIF instead of MyD88, interferon regulatory factor 3 (IRF3) is activated, which leads to the synthesis of type I interferons.



and activation of IRAK4, the initial protein kinase in the TLR signal transduction pathway. This is followed by recruitment of IRAK1 and TRAF-6 to the pathway. This pathway then bifurcates, one branch activating the MAP kinases, the other leading to activation of the transcription factor NF κ B (Fig. 29.4). The steps leading to the activation of NF κ B can be seen in more detail in Fig. 23.1. Activation of the classical NF κ B pathway results in the production of pro-inflammatory cytokines such as IL-6, IL-12, and TNF- α (see Fig. 33.2). A functional IRAK4 is critical to the ability to make responses to TLR ligands, because signaling via the NF κ B pathway is blocked in its absence. In addition to its role in innate immunity, IRAK4 may also be directly involved in the development of optimal adaptive immune responses. Antigen-stimulated T cells from mice with inactivated IRAK4 kinase secrete reduced amounts of IL-17, a cytokine that is important in antibacterial immunity. Recently, mice lacking IRAK4 were found to have reduced splenic and peripheral expansion of CD8 T cells in response to infection with lymphocytic choriomeningitis virus, suggesting that IRAK4 may be required for optimal antiviral CD8 T-cell responses *in vivo*. An important point to note is that IRAK4-deficient mice are susceptible to both bacterial and viral infections, whereas the IRAK4-deficient human patients identified so far seem to be primarily susceptible to pyogenic bacteria. As the adaptor protein MyD88 is upstream of IRAK-4, patients with MyD88 deficiency have a similar presentation as those with IRAK-4 deficiency.

More than 28 patients with IRAK4 deficiency have now been identified as a result of their severe, recurrent infections with pyogenic bacteria. By routine immunological criteria they appear normal, with normal numbers of B and T cells, normal serum immunoglobulin levels, and protective antibody titers to protein antigens, but variably impaired antibody titers to polysaccharide antigens. A longitudinal study of IRAK4-deficient patients has observed that their susceptibility to invasive bacterial infections decreases with age, becoming similar to that of the normal population by 14 years old. The reason for this clinical improvement with age is unknown, but compensation by the adaptive immune system has been hypothesized. To completely understand the role of IRAK4 in human innate and adaptive immunity will require the identification and evaluation of more patients with this defect.

TLR-3 and TLR-4 can also signal via a pathway that does not use IRAK4. They can use a second, MyD88-independent, pathway involving the adaptor proteins TRIF (for TLR-3) or TRIF/TRAM (for TLR-4). This pathway results in the activation of the interferon regulatory factor 3 (IRF3) and the production of type I interferons (see Fig. 29.4). Interferon production is an essential function of the innate response

to certain viruses, including the herpesvirus family and the human immunodeficiency virus (HIV). Patients with TRIF deficiency have a selective defect in their ability to contain HSV replication in non-hematopoietic cells, and are particularly susceptible to HSV encephalitis. Type I interferon production is also stimulated in plasmacytoid dendritic cells and monocytes by the activation of TLR-7, TLR-8, and TLR-9. Like the other TLRs, these receptors can signal via the IRAK/TRAF-6 pathway, leading to the production of pro-inflammatory cytokines. IRAK1 can, however, also phosphorylate and activate the transcription factor IRF7, which induces the expression of the gene encoding IFN- α . Virus-induced activation of IRF7 and the expression of the gene encoding IFN- α production is essential for host defense against viruses. Indeed, subsequent herpes simplex virus infection in mice lacking IRF7 is lethal.

Questions.

- 1 Douglas's immune deficiency is characterized by susceptibility to severe, invasive pneumococcal infection. What other immunodeficiencies are associated with similar susceptibility to pneumococci?
- 2 What clue in the history of a patient with recurrent pneumococcal or staphylococcal infections would favor a possible diagnosis of IRAK4 deficiency?
- 3 What is the clinical course of patients with known defects in IRAK4?
- 4 Why might IRAK-deficient patients fail to respond to immunization with polysaccharide antigens?
- 5 IRAK4 seems to be involved in signaling from all TLRs. Why has increased susceptibility to viral infections not yet been documented in human patients?
- 6 How should IRAK4-deficient patients be managed to prevent the recurrence of invasive infection?